

TOLEDO LIGHT 2, OH, USA

WMO: 997344

Lat: 41.826N

Lon: 83.194W

Elev: 617

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.7	(c) 7.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 40.4	(k) 10.5	(l) 36.9	(m) 20.0	(n) 15.9	(o) 280	(p) 0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 84.9	(d) N/A	(e) 82.8	(f) N/A	(g) 80.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 11.4	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 35.3	(b) 32.0	(c) 28.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	25.4	26.3	34.4	47.6	59.4	69.7	74.3	73.9	67.3	54.9	42.5	34.3	
	DBStd	18.94	9.87	10.07	7.99	6.85	7.04	5.54	4.76	4.13	6.13	7.66	8.34	7.31	
	HDD50	2813	762	664	486	126	6	0	0	0	0	35	247	487	
	HDD65	6007	1227	1084	950	524	202	20	1	1	45	327	675	951	
	CDD50	3165	0	0	1	53	298	590	755	740	520	185	22	1	
	CDD65	885	0	0	0	1	29	159	291	276	116	13	0	0	
	CDH74	3964	0	0	0	1	78	570	1662	1282	348	23	0	0	
	CDH80	562	0	0	0	0	8	78	284	141	48	3	0	0	
Wind	WSAvg	14.9	17.7	15.9	14.7	16.5	13.7	12.8	12.1	12.1	13.8	16.6	17.0	16.3	
Precipitation	PrecAvg	33.3	2.0	1.9	2.4	3.3	3.4	3.5	3.3	3.3	3.1	2.4	2.6	2.3	
	PrecMax	46.9	3.6	5.1	5.5	5.0	5.8	8.0	6.4	7.9	6.2	6.1	6.6	6.0	
	PrecMin	25.3	0.6	0.4	0.6	1.1	0.8	1.7	1.3	0.6	0.6	0.5	0.8	0.6	
	PrecStd	4.9	0.9	1.1	1.1	1.1	1.6	1.4	1.1	1.5	1.4	1.3	1.2	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	53.4	57.2	72.1	80.4	85.8	87.7	86.2	84.9	77.2	63.5	54.6	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	46.4	46.2	51.8	67.2	76.4	82.3	85.4	83.5	80.6	72.4	59.5	50.4	N/A
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	42.2	41.3	47.4	62.2	72.9	79.6	83.5	81.5	77.8	69.0	56.0	46.9	N/A
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	38.5	37.7	43.4	57.9	69.6	77.3	81.3	79.8	75.5	65.7	53.4	43.6	N/A
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.5	10.0	10.4	10.0	9.2	8.8	8.6	9.0	8.7	8.3	7.2	
		MCDBR	11.9	13.5	14.6	18.7	13.8	11.9	10.5	10.2	11.7	10.1	12.1	10.7	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.304	0.335	0.357	0.401	0.436	0.449	0.441	0.433	0.388	0.362	0.322	0.316	
	taud		2.410	2.301	2.328	2.239	2.187	2.201	2.245	2.251	2.354	2.407	2.481	2.408	
	Ebn at Noon		267	273	281	275	267	263	264	263	268	261	259	251	
	Edh at Noon		25	33	36	42	46	45	43	41	35	29	23	23	
All-Sky Solar Radiation	RadAvg		473	752	1103	1465	1719	1922	1929	1686	1365	895	565	389	
	RadStd		49	97	102	122	147	127	104	64	114	73	77	37	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	N/A	(47)

Nomenclature: See separate page